

Towhid Ahmed

RAG Engineer

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Professional Summary

RAG Engineer with 1+ year of freelance experience building production-grade LLM applications, multi-agent systems, and RAG solutions for SaaS and e-commerce platforms. Developed 3 AI applications using LangChain, LangGraph, and FastAPI, achieving 35% hallucination reduction, 25% retrieval relevance improvement, and 98/98 passing automated tests. Experienced in RAG pipelines, LLMops, Docker, and LangSmith.

Technical Skills

Programming: Python, SQL

RAG & Retrieval: RAG, Advanced RAG, Semantic Search, Vector Search, Cross-Encoder Reranking, Pinecone

LLMs & Generative AI: LLMs, Llama, OpenAI API, Groq, Prompt Engineering, AI Agents

AI Frameworks: LangChain, LangGraph, Hugging Face Transformers, Sentence Transformers

Vector Databases: Pinecone, ChromaDB, PostgreSQL

Evaluation & LLMops: LangSmith, LLM Evaluation, LLM Guardrails, RAG Evaluation

Backend & Deployment: FastAPI, REST APIs, Async Programming, Docker

Professional Experience

Freelance Generative AI Engineer | Self-Employed | Jun 2025 – Present

AI Document Assistant | Jun 2026 – Jul 2026

- Enhanced information reliability and user trust by developing a 3-agent LangGraph workflow with citation-based RAG, significantly reducing AI hallucinations.
- Powered seamless document processing and interaction by engineering 7 production-ready REST APIs using FastAPI, OpenAI API, and Pinecone.
- Enabled 1,000+ document knowledge-base updates without redeployment.

AI Sales & Customer Support Assistant | Aug 2025 – Sep 2025

- Automated customer interactions and improved consistency using a multi-agent LangGraph pipeline (Intent, Retrieval, Response) with Pinecone RAG and conversational memory.
- Slashed AI inference costs and maintained high execution speed by integrating Groq, Hugging Face embeddings, and Pinecone Serverless.
- Built a scalable support infrastructure with a Streamlit UI and a FastAPI backend featuring 6+ REST APIs for knowledge uploads and session management.

Selected Projects

Multi-Agent AI System — Python, LangGraph, LangChain, FastAPI, Pinecone, LangSmith, Streamlit, Docker, Render, Hugging Face, Tavily GitHub | Demo

- Designed 4-agent LangGraph workflow routing queries between RAG and web search based on intent and complexity.
- Built FastAPI backend with async execution, 7 core APIs (/chat, /upload, /health, /metrics) optimized for Render.
- Integrated Tavily with DuckDuckGo fallback and LangSmith tracing for reliable web retrieval and observability.

Multimodal RAG — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, Git, GitHub, REST API, Async Programming, Caching GitHub

- Designed production multimodal retrieval (PDF/audio/video) improving relevance by 25% and reducing hallucinations by 35%.
- Implemented two-stage retrieval (semantic chunking + cross-encoder re-ranking) with <2s latency via caching.
- Tech: Groq Llama 3.3, BAAI embeddings, Pinecone, FastAPI, Streamlit, Docker; 76-unit test coverage.

Clinical RAG Assistant — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT GitHub

- Built a clinical information assistant using a 4-agent LangGraph pipeline (Router, Safety, Retrieval, Answer) with RAG and web search.
- Implemented multi-layer safety system (input filters, retrieval validation, output sanitization) with JWT auth.
- Achieved 98/98 passing tests across guardrails, RAG evaluation, and API workflows.

Education and Certificates

Education: Computer Science Engineering(CSE), Chittagong University of Engineering & Technology | 2023-2027

Certifications: • Generative AI & Agentic AI — Krish Naik (Hands-on LLM systems & agents)

• LLMops — Udemy (Deployment, monitoring, CI/CD for LLM apps) • Data Structures & Algorithms — TakeUForward